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Faculty of Forestry &
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Dear Dr. Way:

Please consider our paper, entitled ‘The problem and promise of modeling plant leafout under warmer winters,’ as a perspective for *Global Change Biology*.

What scientific question is addressed in this manuscript?

Why do current models of leafout timing with climate change make wildly different predictions for the same system and temperature regime? How can simple models from the 1970s still be equally predictive as newer models, despite apparent progress over the last decades?

What is/are the key finding(s) that answer this question?

We argue progress has stalled because we do not understand how winter temperatures affect leafout. Models today estimate the effect of winter temperatures via an unobserved process called chilling. This has produced fundamental problems that the field has struggled to admit or address.

Why is this work important and timely?

Interest in chilling for forecasting the effects of climate change has led to myriad new models, but few insights. In this perspective we argue that major advances through molecular and other new approaches will be slow without a reckoning with the concept of chilling.

Describe how your paper fits within the scope of GCB.

We bring a cross-disciplinary and solutions-focused perspective on chilling, a concept critical to predicting and mitigating impacts of climate change. We bridge across ecological forecasting, molecular biology, forest tree and crop modelers to highlight the problems that may limit the potential for rapid progress, and show how to fix them.

What biological AND global change aspects does it address?

Chilling is a fundamental concept in plant biology with major implications for many fruit crops and how well forests can continue to sequester carbon with climate change. An improved understanding of chilling is integral to managing agricultural, horticultural, and forest systems with continued climate change.

What are the three most recently published papers that are relevant to this manuscript?

Baumgarten, F., Zohner, C.M., Gessler, A. and Vitasse, Y. (2021), Chilled to be forced: the best dose to wake up buds from winter dormancy. *New Phytologist* 230: 1366-1377.

Ettinger, A.K., Chamberlain, C.J., Morales-Castilla, I., Buonaiuto, D.M., Flynn, D.F.B., Savas, T., Samaha, J.A. & Wolkovich, E.M. (2020) Winter temperatures predominate in spring phenological responses to warming. *Nature Climate Change* 10, 1137–U119.

Zhao, Y., Antoniou-Kourounioti, R.L., Calder, G., Dean, C. & Howard, M. (2020) Temperature-dependent growth contributes to long-term cold sensing. *Nature* 583, 825–829.

Provide the name and institutional email addresses of 5 potential reviewers.

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I and my co-authors believe this perspective would be of interest to readers *Global Change Biology* and look forward to hearing from you. If have additional questions, please let me know.

Sincerely,

A handwritten signature in black ink, appearing to read 'Elizabeth Wolkovich', with a stylized flourish at the end.

Elizabeth M Wolkovich
Associate Professor of Forest & Conservation Sciences